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Dynamic allocation of a capacitive component in a memory device

Abstract

Methods and devices for dynamic allocation of a capacitive component in a memory device are described. A memory device may include one or more voltage rails for distributing supply voltages to a memory die. A memory device may include a capacitive component that may be dynamically coupled to a voltage rail based on an identification of an operating condition on the memory die, such as a voltage droop on the voltage rail. The capacitive component may be dynamically coupled with the voltage rail to maintain the supply voltage on the voltage rail during periods of high demand. The capacitive component may be dynamically switched between voltage rails during operation of the memory device based on operating conditions associated with the voltage rails.

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Background/Summary

CROSS REFERENCE (1) The present application for patent is a continuation of U.S. patent application Ser. No. 17/569,303 by Badrieh et al., entitled "DYNAMIC ALLOCATION OF A CAPACITIVE COMPONENT IN A MEMORY DEVICE," filed Jan. 5, 2022, which is a continuation of U.S. patent application Ser. No. 17/025,628 by Badrieh et al., entitled "DYNAMIC ALLOCATION OF A CAPACITIVE COMPONENT IN A MEMORY DEVICE," filed Sep. 18, 2020, now U.S. Pat. No. 11,238,903, which is a continuation of U.S. patent application Ser. No. 16/268,092 by Badrieh et al., entitled "DYNAMIC ALLOCATION OF A CAPACITIVE COMPONENT IN A MEMORY DEVICE," filed Feb. 5, 2019, now U.S. Pat. No. 10,796,729, each of which assigned to the assignee hereof and each of which is expressly incorporated by reference in its entirety herein.

BACKGROUND

(1) The following relates generally to a system that includes at least one memory device and more specifically to dynamic allocation of a capacitive component in a memory device.

(2) Memory devices are widely used to store information in various electronic devices such as computers, wireless communication devices, cameras, digital displays, and the like. Information is stored by programing different states of a memory device. For example, binary devices most often store one of two states, often denoted by a logic 1 or a logic 0. In other devices, more than two states may be stored. To access the stored information, a component of the device may read, or sense, at least one stored state in the memory device. To store information, a component of the device may write, or program, the state in the memory device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of a system that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein.

(2) FIG. 2 illustrates an example of a memory die that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein.

(3) FIG. 3 illustrates an example of a circuit that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein.

(4) FIG. 4 illustrates an example of a circuit that supports dynamic allocation of a capacitive

component in a memory device in accordance with examples as disclosed herein.

(5) FIG. 5 illustrates an example of a circuit that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein.

(6) FIG. 6 shows a block diagram of a memory device that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein.

(7) FIGS. 7 and 8 show flowcharts illustrating a method or methods that support dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein.

DETAILED DESCRIPTION

(8) A memory die may include a power distribution network (PDN) to distribute power to various components of the memory die. The PDN may receive electrical power from one or more power supplies or voltage sources and may distribute the power or voltages across the die using one or more voltage “rails” (e.g., conductive paths). For example, a memory die may include a V_{sub}.DD voltage rail for distributing a V_{sub}.DD supply voltage to the memory die, a V_{sub}.SS voltage rail for distributing a V_{sub}.SS supply voltage to the memory die, etc. The voltage rails may each be coupled with or include a network of resistors separated by conductive lines, which may be referred to as a resistive grid.

(9) In some cases, a voltage along a voltage rail may droop (e.g., drop, decrease) or spike (e.g., rise, increase) relative to the corresponding supply voltage. For example, the voltage on the voltage rail may droop in response to the voltage or current demand of various components on the memory die, such as memory cells, switches, amplifiers, etc. In some cases, if a voltage rail experiences a relatively large droop, the rail may be unable to provide sufficient voltage or current to components on the memory die to enable proper operation.

(10) A resistive grid in a PDN may include various capacitive elements (e.g., capacitors) coupled with the corresponding voltage rail to help the voltage rail maintain the supply voltage over a range of operating conditions. Such capacitive elements may provide additional charge (e.g., voltage) to a voltage rail over short durations of high demand, and thus their inclusion may be conceptually similar to having small auxiliary power supplies distributed throughout the PDN.

(11) In some cases, the size or quantity of capacitors coupled with a voltage rail may affect the range of voltage variations on the rail that can be compensated to maintain the supply voltage along the rail. That is, a larger amount of capacitance may be able to provide (or absorb) more charge to/from a voltage rail, and therefore may be able to compensate for a larger range of voltage variations on the rail. Using larger or more numerous capacitors, however, may consume more area and may cost more than using smaller or fewer capacitors. Thus, in some cases, the size and/or quantity of capacitors included in a PDN may be a design consideration for the PDN.

(12) In some cases, a circuit may be designed to use a relatively large amount of capacitance to compensate for worst-case voltage variations on a voltage rail. Such large capacitance may not be needed under most operating conditions, and therefore may represent wasted resources during periods when the additional capacitance is not needed.

(13) A memory die may include a capacitive component (such as one or more capacitors) that may be dynamically coupled with a voltage rail to compensate for voltage variations on the rail when certain operating conditions are identified, such as a relatively large voltage droop or spike. The operating conditions may be determined by a control component based on, for example, sensing that a voltage on the memory die satisfies a threshold condition, or predicting that a voltage on the memory die will satisfy a threshold condition based on a received sequence of commands.

(14) In some cases, the capacitive component may be switched between a first voltage rail and a second voltage rail based on operating conditions associated with the first rail and/or the second rail. That is, in some cases, the capacitive component may be dynamically allocated to whichever voltage rail is experiencing excessive demand.

(15) As previously discussed, each voltage rail may be associated with a resistive grid that includes

discrete capacitors (e.g., separate from the dynamically allocated capacitive component). In some cases, instead of or in addition to coupling a dynamically allocated capacitive component with the voltage rails, two voltage rails associated with the same supply voltage may be dynamically coupled with each other to provide additional capacitance to the voltage rails during periods of high demand.

(16) Features of the disclosure are initially described in the context of a memory system and memory die as described with reference to FIGS. 1 and 2. Features of the disclosure are described in the context of a memory die with multiple voltage rails as described with reference to FIGS. 3-5. These and other features of the disclosure are further illustrated by and described with reference to an apparatus diagram and flowcharts that relate to dynamic allocation of a capacitive component in a memory device as described with reference to FIGS. 6-8.

(17) FIG. 1 illustrates an example of a system **100** that utilizes one or more memory devices in accordance with examples as disclosed herein. The system **100** may include an external memory controller **105**, a memory device **110**, and a plurality of channels **115** coupling the external memory controller **105** with the memory device **110**. The system **100** may include one or more memory devices, but for ease of description the one or more memory devices may be described as a single memory device **110**.

(18) The system **100** may include aspects of an electronic device, such as a computing device, a mobile computing device, a wireless device, or a graphics processing device. The system **100** may be an example of a portable electronic device. The system **100** may be an example of a computer, a laptop computer, a tablet computer, a smartphone, a cellular phone, a wearable device, an internet-connected device, or the like. The memory device **110** may be component of the system configured to store data for one or more other components of the system **100**. In some examples, the system **100** is configured for bi-directional wireless communication with other systems or devices using a base station or access point. In some examples, the system **100** is capable of machine-type communication (MTC), machine-to-machine (M2M) communication, or device-to-device (D2D) communication.

(19) At least portions of the system **100** may be examples of a host device. Such a host device may be an example of a device that uses memory to execute processes such as a computing device, a mobile computing device, a wireless device, a graphics processing device, a computer, a laptop computer, a tablet computer, a smartphone, a cellular phone, a wearable device, an internet-connected device, some other stationary or portable electronic device, or the like. In some cases, the host device may refer to the hardware, firmware, software, or a combination thereof that implements the functions of the external memory controller **105**. In some cases, the external memory controller **105** may be referred to as a host or host device. In some examples, system **100** is a graphics card.

(20) In some cases, a memory device **110** may be an independent device or component that is configured to be in communication with other components of the system **100** and provide physical memory addresses/space to potentially be used or referenced by the system **100**. In some examples, a memory device **110** may be configurable to work with at least one or a plurality of different types of systems **100**. Signaling between the components of the system **100** and the memory device **110** may be operable to support modulation schemes to modulate the signals, different pin designs for communicating the signals, distinct packaging of the system **100** and the memory device **110**, clock signaling and synchronization between the system **100** and the memory device **110**, timing conventions, and/or other factors.

(21) The memory device **110** may be configured to store data for the components of the system **100**. In some cases, the memory device **110** may act as a slave-type device to the system **100** (e.g., responding to and executing commands provided by the system **100** through the external memory controller **105**). Such commands may include an access command for an access operation, such as a write command for a write operation, a read command for a read operation, a refresh command

for a refresh operation, or other commands. The memory device **110** may include two or more memory dice **160** (e.g., memory chips) to support a desired or specified capacity for data storage. The memory device **110** including two or more memory dice may be referred to as a multi-die memory or package (also referred to as multi-chip memory or package).

(22) The system **100** may further include a processor **120**, a basic input/output system (BIOS) component **125**, one or more peripheral components **130**, and an input/output (I/O) controller **135**. The components of system **100** may be in electronic communication with one another using a bus **140**.

(23) The processor **120** may be configured to control at least portions of the system **100**. The processor **120** may be a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or it may be a combination of these types of components. In such cases, the processor **120** may be an example of a central processing unit (CPU), a graphics processing unit (GPU), a general purpose GPU (GPGPU), or a system on a chip (SoC), among other examples.

(24) The BIOS component **125** may be a software component that includes a BIOS operated as firmware, which may initialize and run various hardware components of the system **100**. The BIOS component **125** may also manage data flow between the processor **120** and the various components of the system **100**, e.g., the peripheral components **130**, the I/O controller **135**, etc. The BIOS component **125** may include a program or software stored in read-only memory (ROM), flash memory, or any other non-volatile memory.

(25) The peripheral component(s) **130** may be any input device or output device, or an interface for such devices, that may be integrated into or with the system **100**. Examples may include disk controllers, sound controller, graphics controller, Ethernet controller, modem, universal serial bus (USB) controller, a serial or parallel port, or peripheral card slots, such as peripheral component interconnect (PCI) or specialized graphics ports. The peripheral component(s) **130** may be other components understood by those skilled in the art as peripherals.

(26) The I/O controller **135** may manage data communication between the processor **120** and the peripheral component(s) **130**, input devices **145**, or output devices **150**. The I/O controller **135** may manage peripherals that are not integrated into or with the system **100**. In some cases, the I/O controller **135** may represent a physical connection or port to external peripheral components.

(27) The input **145** may represent a device or signal external to the system **100** that provides information, signals, or data to the system **100** or its components. This may include a user interface or interface with or between other devices. In some cases, the input **145** may be a peripheral that interfaces with system **100** via one or more peripheral components **130** or may be managed by the I/O controller **135**.

(28) The output **150** may represent a device or signal external to the system **100** configured to receive an output from the system **100** or any of its components. Examples of the output **150** may include a display, audio speakers, a printing device, or another processor on printed circuit board, and so forth. In some cases, the output **150** may be a peripheral that interfaces with the system **100** via one or more peripheral components **130** or may be managed by the I/O controller **135**.

(29) The components of system **100** may be made up of general-purpose or special purpose circuitry designed to carry out their functions. This may include various circuit elements, for example, conductive lines, transistors, capacitors, inductors, resistors, amplifiers, or other active or passive elements, configured to carry out the functions described herein.

(30) The memory device **110** may include a device memory controller **155** and one or more memory dice **160**. Each memory die **160** may include a local memory controller **165** (e.g., local memory controller **165-a**, local memory controller **165-b**, and/or local memory controller **165-N**) and a memory array **170** (e.g., memory array **170-a**, memory array **170-b**, and/or memory array **170-N**). A memory array **170** may be a collection (e.g., a grid) of memory cells, with each memory

cell being configured to store at least one bit of digital data. Features of memory arrays **170** and/or memory cells are described in more detail with reference to FIG. 2.

(31) In some cases, a memory device **110** may include a power distribution network (PDN) for distributing power received from a power supply (e.g., a voltage supply, voltage regulator) to memory array **170**. The PDN may include one or more voltage rails, each of which may be coupled with a voltage supply. In some cases, the PDN may include a capacitive component that may be dynamically switched between voltage rails to provide additional capacitance (e.g., additional charge) to the rail during periods of high demand and thereby maintain the supply voltage on the voltage rail.

(32) The memory device **110** may be an example of a two-dimensional (2D) array of memory cells or may be an example of a three-dimensional (3D) array of memory cells. For example, a 2D memory device may include a single memory die **160**. A 3D memory device may include two or more memory dice **160** (e.g., memory die **160-a**, memory die **160-b**, and/or any quantity of memory dice **160-N**). In a 3D memory device, a plurality of memory dice **160-N** may be stacked on top of one another or next to one another. In some cases, memory dice **160-N** in a 3D memory device may be referred to as decks, levels, layers, or dies. A 3D memory device may include any quantity of stacked memory dice **160-N** (e.g., two high, three high, four high, five high, six high, seven high, eight high). This may increase the quantity of memory cells that may be positioned on a substrate as compared with a single 2D memory device, which in turn may reduce production costs or increase the performance of the memory array, or both. In some 3D memory device, different decks may share at least one common access line such that some decks may share at least one of a word line, a digit line, and/or a plate line.

(33) The device memory controller **155** may include circuits or components configured to control operation of the memory device **110**. As such, the device memory controller **155** may include the hardware, firmware, and software that enables the memory device **110** to perform commands and may be configured to receive, transmit, or execute commands, data, or control information related to the memory device **110**. The device memory controller **155** may be configured to communicate with the external memory controller **105**, the one or more memory dice **160**, or the processor **120**. In some cases, the memory device **110** may receive data and/or commands from the external memory controller **105**. For example, the memory device **110** may receive a write command indicating that the memory device **110** is to store certain data on behalf of a component of the system **100** (e.g., the processor **120**) or a read command indicating that the memory device **110** is to provide certain data stored in a memory die **160** to a component of the system **100** (e.g., the processor **120**). In some cases, the device memory controller **155** may control operation of the memory device **110** described herein in conjunction with the local memory controller **165** of the memory die **160**. Examples of the components included in the device memory controller **155** and/or the local memory controllers **165** may include receivers for demodulating signals received from the external memory controller **105**, decoders for modulating and transmitting signals to the external memory controller **105**, logic, decoders, amplifiers, filters, or the like.

(34) The local memory controller **165** (e.g., local to a memory die **160**) may be configured to control operations of the memory die **160**. Also, the local memory controller **165** may be configured to communicate (e.g., receive and transmit data and/or commands) with the device memory controller **155**. The local memory controller **165** may support the device memory controller **155** to control operation of the memory device **110** as described herein. In some cases, the memory device **110** does not include the device memory controller **155**, and the local memory controller **165** or the external memory controller **105** may perform the various functions described herein. As such, the local memory controller **165** may be configured to communicate with the device memory controller **155**, with other local memory controllers **165**, or directly with the external memory controller **105** or the processor **120**.

(35) The external memory controller **105** may be configured to enable communication of

information, data, and/or commands between components of the system **100** (e.g., the processor **120**) and the memory device **110**. The external memory controller **105** may act as a liaison between the components of the system **100** and the memory device **110** so that the components of the system **100** may not need to know the details of the memory device's operation. The components of the system **100** may present requests to the external memory controller **105** (e.g., read commands or write commands) that the external memory controller **105** satisfies. The external memory controller **105** may convert or translate communications exchanged between the components of the system **100** and the memory device **110**. In some cases, the external memory controller **105** may include a system clock that generates a common (source) system clock signal. In some cases, the external memory controller **105** may include a common data clock that generates a common (source) data clock signal.

(36) In some cases, the external memory controller **105** or other component of the system **100**, or its functions described herein, may be implemented by the processor **120**. For example, the external memory controller **105** may be hardware, firmware, or software, or some combination thereof implemented by the processor **120** or other component of the system **100**. While the external memory controller **105** is depicted as being external to the memory device **110**, in some cases, the external memory controller **105**, or its functions described herein, may be implemented by a memory device **110**. For example, the external memory controller **105** may be hardware, firmware, or software, or some combination thereof implemented by the device memory controller **155** or one or more local memory controllers **165**. In some cases, the external memory controller **105** may be distributed across the processor **120** and the memory device **110** such that portions of the external memory controller **105** are implemented by the processor **120** and other portions are implemented by a device memory controller **155** or a local memory controller **165**. Likewise, in some cases, one or more functions ascribed herein to the device memory controller **155** or local memory controller **165** may in some cases be performed by the external memory controller **105** (either separate from or as included in the processor **120**).

(37) A controller (e.g., external memory controller **105**, device memory controller **155**, and/or local memory controller **165**, or combination thereof) may identify an operating condition associated with a voltage rail in the PDN (e.g., a measured or predicted voltage droop or spike), and may respond by coupling a capacitive component with the voltage rail to maintain the voltage on the voltage rail. In some cases, the controller may predict a voltage droop or spike on a voltage rail based on receiving a sequence of one or more commands that specify memory operations that may when executed sequentially or in parallel, cause the voltage on a voltage rail to droop or spike.

(38) The components of the system **100** may exchange information with the memory device **110** using a plurality of channels **115**. In some examples, the channels **115** may enable communications between the external memory controller **105** and the memory device **110**. Each channel **115** may include one or more signal paths or transmission mediums (e.g., conductors) between terminals associated with the components of system **100**. For example, a channel **115** may include a first terminal including one or more pins or pads at external memory controller **105** and one or more pins or pads at the memory device **110**. A pin may be an example of a conductive input or output point of a device of the system **100**, and a pin may be configured to act as part of a channel.

(39) In some cases, a pin or pad of a terminal may be part of to a signal path of the channel **115**. Additional signal paths may be coupled with a terminal of a channel for routing signals within a component of the system **100**. For example, the memory device **110** may include signal paths (e.g., signal paths internal to the memory device **110** or its components, such as internal to a memory die **160**) that route a signal from a terminal of a channel **115** to the various components of the memory device **110** (e.g., a device memory controller **155**, memory dice **160**, local memory controllers **165**, memory arrays **170**).

(40) Channels **115** (and associated signal paths and terminals) may be dedicated to communicating specific types of information. In some cases, a channel **115** may be an aggregated channel and thus

may include multiple individual channels. For example, a data channel **190** may be x4 (e.g., including four signal paths), x8 (e.g., including eight signal paths), x16 (including sixteen signal paths), and so forth. Signals communicated over the channels may use double data rate (DDR) signaling. For example, some symbols of a signal may be registered on a rising edge of a clock signal and other symbols of the signal may be registered on a falling edge of the clock signal. Signals communicated over channels may use single data rate (SDR) signaling. For example, one symbol of the signal may be registered for each clock cycle.

(41) In some cases, the channels **115** may include one or more command and address (CA) channels **186**. The CA channels **186** may be configured to communicate commands between the external memory controller **105** and the memory device **110** including control information associated with the commands (e.g., address information). For example, the CA channel **186** may include a read command with an address of the desired data. In some cases, the CA channels **186** may be registered on a rising clock signal edge and/or a falling clock signal edge. In some cases, a CA channel **186** may include any quantity of signal paths to decode address and command data (e.g., eight or nine signal paths).

(42) In some cases, the channels **115** may include one or more clock signal (CK) channels **188**. The CK channels **188** may be configured to communicate one or more common clock signals between the external memory controller **105** and the memory device **110**. Each clock signal may be configured to oscillate between a high state and a low state and coordinate the actions of the external memory controller **105** and the memory device **110**. In some cases, the clock signal may be a differential output (e.g., a CK_t signal and a CK_c signal) and the signal paths of the CK channels **188** may be configured accordingly. In some cases, the clock signal may be single ended. A CK channel **188** may include any quantity of signal paths. In some cases, the clock signal CK (e.g., a CK_t signal and a CK_c signal) may provide a timing reference for command and addressing operations for the memory device **110**, or other system-wide operations for the memory device **110**. The clock signal CK therefore may be variously referred to as a control clock signal CK, a command clock signal CK, or a system clock signal CK. The system clock signal CK may be generated by a system clock, which may include one or more hardware components (e.g., oscillators, crystals, logic gates, transistors, or the like).

(43) In some cases, the channels **115** may include one or more data (DQ) channels **190**. The data channels **190** may be configured to communicate data and/or control information between the external memory controller **105** and the memory device **110**. For example, the data channels **190** may communicate information (e.g., bi-directional) to be written to the memory device **110** or information read from the memory device **110**. The data channels **190** may communicate signals that may be modulated using a variety of different modulation schemes (e.g., NRZ, PAM4).

(44) In some cases, the channels **115** may include one or more other channels **192** that may be dedicated to other purposes. These other channels **192** may include any quantity of signal paths.

(45) The channels **115** may couple the external memory controller **105** with the memory device **110** using a variety of different architectures. Examples of the various architectures may include a bus, a point-to-point connection, a crossbar, a high-density interposer such as a silicon interposer, or channels formed in an organic substrate or some combination thereof. For example, in some cases, the signal paths may at least partially include a high-density interposer, such as a silicon interposer or a glass interposer.

(46) Signals communicated over the channels **115** may be modulated using a variety of different modulation schemes. In some cases, a binary-symbol (or binary-level) modulation scheme may be used to modulate signals communicated between the external memory controller **105** and the memory device **110**. A binary-symbol modulation scheme may be an example of a M-ary modulation scheme where M is equal to two. Each symbol of a binary-symbol modulation scheme may be configured to represent one bit of digital data (e.g., a symbol may represent a logic 1 or a logic 0). Examples of binary-symbol modulation schemes include, but are not limited to, non-

return-to-zero (NRZ), unipolar encoding, bipolar encoding, Manchester encoding, pulse amplitude modulation (PAM) having two symbols (e.g., PAM2), and/or others.

(47) In some cases, a multi-symbol (or multi-level) modulation scheme may be used to modulate signals communicated between the external memory controller **105** and the memory device **110**. A multi-symbol modulation scheme may be an example of a M-ary modulation scheme where M is greater than or equal to three. Each symbol of a multi-symbol modulation scheme may be configured to represent more than one bit of digital data (e.g., a symbol may represent a logic 00, a logic 01, a logic 10, or a logic 11). Examples of multi-symbol modulation schemes include, but are not limited to, PAM4, PAM8, etc., quadrature amplitude modulation (QAM), quadrature phase shift keying (QPSK), and/or others. A multi-symbol signal or a PAM4 signal may be a signal that is modulated using a modulation scheme that includes at least three levels to encode more than one bit of information. Multi-symbol modulation schemes and symbols may alternatively be referred to as non-binary, multi-bit, or higher-order modulation schemes and symbols.

(48) FIG. 2 illustrates an example of a memory die **200** in accordance with examples as disclosed herein. The memory die **200** may be an example of the memory dice **160** described with reference to FIG. 1. In some cases, the memory die **200** may be referred to as a memory chip, a memory device, or an electronic memory apparatus. The memory die **200** may include one or more memory cells **205** that are programmable to store different logic states. Each memory cell **205** may be programmable to store two or more states. For example, the memory cell **205** may be configured to store one bit of digital logic at a time (e.g., a logic 0 and a logic 1). In some cases, a single memory cell **205** (e.g., a multi-level memory cell) may be configured to store more than one bit of digit logic at a time (e.g., a logic 00, logic 01, logic 10, or a logic 11).

(49) A memory cell **205** may store a charge representative of the programmable states in a capacitor. For example, memory cells in DRAM architectures may each include a capacitor that includes a dielectric material to store a charge representative of the programmable state. In other types of memory architectures, other storage devices and components are possible. For example, nonlinear dielectric materials may be employed.

(50) Operations such as reading and writing may be performed on memory cells **205** by activating or selecting access lines such as a word line **210** and/or a digit line **215**. In some cases, digit lines **215** may also be referred to as bit lines. References to access lines, word lines and digit lines, or their analogues, are interchangeable without loss of understanding or operation. Activating or selecting a word line **210** or a digit line **215** may include applying a voltage to the respective line.

(51) The memory die **200** may include the access lines (e.g., the word lines **210** and the digit lines **215**) arranged in a grid-like pattern. Memory cells **205** may be positioned at intersections of the word lines **210** and the digit lines **215**. By biasing a word line **210** and a digit line **215** (e.g., applying a voltage to the word line **210** or the digit line **215**), a single memory cell **205** may be accessed at their intersection.

(52) Accessing the memory cells **205** may be controlled through a row decoder **220** or a column decoder **225**. For example, a row decoder **220** may receive a row address from the local memory controller **260** and activate a word line **210** based on the received row address. A column decoder **225** may receive a column address from the local memory controller **260** and may activate a digit line **215** based on the received column address. For example, the memory die **200** may include multiple word lines **210**, labeled WL_1 through WL_M, and multiple digit lines **215**, labeled DL_1 through DL_N, where M and N depend on the size of the memory array. Thus, by activating a word line **210** and a digit line **215**, e.g., WL_1 and DL_3, the memory cell **205** at their intersection may be accessed. The intersection of a word line **210** and a digit line **215**, in either a two-dimensional or three-dimensional configuration, may be referred to as an address of a memory cell **205**.

(53) The memory cell **205** may include a logic storage component, such as capacitor **230** and a switching component **235**. The capacitor **230** may be an example of a dielectric capacitor or a ferroelectric capacitor. A first node of the capacitor **230** may be coupled with the switching

component **235** and a second node of the capacitor **230** may be coupled with a voltage source **240**. In some cases, the voltage source **240** may be the cell plate reference voltage, such as V_{pl} , or may be ground, such as V_{ss} . In some cases, the voltage source **240** may be an example of a plate line coupled with a plate line driver. The switching component **235** may be an example of a transistor or any other type of switch device that selectively establishes or de-establishes electronic communication between two components.

(54) Selecting or deselecting the memory cell **205** may be accomplished by activating or deactivating the switching component **235**. The capacitor **230** may be in electronic communication with the digit line **215** using the switching component **235**. For example, the capacitor **230** may be isolated from digit line **215** when the switching component **235** is deactivated, and the capacitor **230** may be coupled with digit line **215** when the switching component **235** is activated. In some cases, the switching component **235** is a transistor and its operation may be controlled by applying a voltage to the transistor gate, where the voltage differential between the transistor gate and transistor source may be greater or less than a threshold voltage of the transistor. In some cases, the switching component **235** may be a p-type transistor or an n-type transistor. The word line **210** may be in electronic communication with the gate of the switching component **235** and may activate/deactivate the switching component **235** based on a voltage being applied to word line **210**.

(55) A word line **210** may be a conductive line in electronic communication with a memory cell **205** that is used to perform access operations on the memory cell **205**. In some architectures, the word line **210** may be in electronic communication with a gate of a switching component **235** of a memory cell **205** and may be configured to control the switching component **235** of the memory cell. In some architectures, the word line **210** may be in electronic communication with a node of the capacitor of the memory cell **205** and the memory cell **205** may not include a switching component.

(56) A digit line **215** may be a conductive line that connects the memory cell **205** with a sense component **245**. In some architectures, the memory cell **205** may be selectively coupled with the digit line **215** during portions of an access operation. For example, the word line **210** and the switching component **235** of the memory cell **205** may be configured to couple and/or isolate the capacitor **230** of the memory cell **205** and the digit line **215**. In some architectures, the memory cell **205** may be in electronic communication (e.g., constant) with the digit line **215**.

(57) The sense component **245** may be configured to detect a state (e.g., a charge) stored on the capacitor **230** of the memory cell **205** and determine a logic state of the memory cell **205** based on the stored state. The charge stored by a memory cell **205** may be extremely small, in some cases. As such, the sense component **245** may include one or more sense amplifiers to amplify the signal output by the memory cell **205**. The sense amplifiers may detect small changes in the charge of a digit line **215** during a read operation and may produce signals corresponding to a logic state 0 or a logic state 1 based on the detected charge. During a read operation, the capacitor **230** of memory cell **205** may output a signal (e.g., discharge a charge) to its corresponding digit line **215**. The signal may cause a voltage of the digit line **215** to change. The sense component **245** may be configured to compare the signal received from the memory cell **205** across the digit line **215** to a reference signal **250** (e.g., reference voltage). The sense component **245** may determine the stored state of the memory cell **205** based on the comparison. For example, in binary-signaling, if digit line **215** has a higher voltage than the reference signal **250**, the sense component **245** may determine that the stored state of memory cell **205** is a logic 1 and, if the digit line **215** has a lower voltage than the reference signal **250**, the sense component **245** may determine that the stored state of the memory cell **205** is a logic 0. The sense component **245** may include various transistors or amplifiers to detect and amplify a difference in the signals. The detected logic state of memory cell **205** may be output through column decoder **225** as output **255**. In some cases, the sense component **245** may be part of another component (e.g., a column decoder **225**, row decoder **220**). In some cases, the sense component **245** may be in electronic communication with the row decoder **220** or

the column decoder **225**.

(58) The local memory controller **260** may control the operation of memory cells **205** through the various components (e.g., row decoder **220**, column decoder **225**, and sense component **245**). The local memory controller **260** may be an example of the local memory controller **165** described with reference to FIG. **1**. In some cases, one or more of the row decoder **220**, column decoder **225**, and sense component **245** may be co-located with the local memory controller **260**. The local memory controller **260** may be configured to receive commands and/or data from an external memory controller **105** (or a device memory controller **155** described with reference to FIG. **1**), translate the commands and/or data into information that can be used by the memory die **200**, perform one or more operations on the memory die **200**, and communicate data from the memory die **200** to the external memory controller **105** (or the device memory controller **155**) in response to performing the one or more operations. The local memory controller **260** may generate row and column address signals to activate the target word line **210** and the target digit line **215**. The local memory controller **260** may also generate and control various voltages or currents used during the operation of the memory die **200**. In general, the amplitude, shape, or duration of an applied voltage or current discussed herein may be adjusted or varied and may be different for the various operations discussed in operating the memory die **200**.

(59) In some cases, the local memory controller **260** may be configured to perform a write operation (e.g., a programming operation) on one or more memory cells **205** of the memory die **200**. During a write operation, a memory cell **205** of the memory die **200** may be programmed to store a desired logic state. In some cases, a plurality of memory cells **205** may be programmed during a single write operation. The local memory controller **260** may identify a target memory cell **205** on which to perform the write operation. The local memory controller **260** may identify a target word line **210** and a target digit line **215** in electronic communication with the target memory cell **205** (e.g., the address of the target memory cell **205**). The local memory controller **260** may activate the target word line **210** and the target digit line **215** (e.g., applying a voltage to the word line **210** or digit line **215**), to access the target memory cell **205**. The local memory controller **260** may apply a specific signal (e.g., voltage) to the digit line **215** during the write operation to store a specific state (e.g., charge) in the capacitor **230** of the memory cell **205**, the specific state (e.g., charge) may be indicative of a desired logic state.

(60) In some cases, the local memory controller **260** may be configured to perform a read operation (e.g., a sense operation) on one or more memory cells **205** of the memory die **200**. During a read operation, the logic state stored in a memory cell **205** of the memory die **200** may be determined. In some cases, a plurality of memory cells **205** may be sensed during a single read operation. The local memory controller **260** may identify a target memory cell **205** on which to perform the read operation. The local memory controller **260** may identify a target word line **210** and a target digit line **215** in electronic communication with the target memory cell **205** (e.g., the address of the target memory cell **205**). The local memory controller **260** may activate the target word line **210** and the target digit line **215** (e.g., applying a voltage to the word line **210** or digit line **215**), to access the target memory cell **205**. The target memory cell **205** may transfer a signal to the sense component **245** in response to biasing the access lines. The sense component **245** may amplify the signal. The local memory controller **260** may fire the sense component **245** (e.g., latch the sense component) and thereby compare the signal received from the memory cell **205** to the reference signal **250**. Based on that comparison, the sense component **245** may determine a logic state that is stored on the memory cell **205**. The local memory controller **260** may communicate the logic state stored on the memory cell **205** to the external memory controller **105** (or the device memory controller **155**) as part of the read operation.

(61) Memory die **200** may include or be coupled with one or more voltage rails that may supply power to some or all of memory die **200**. In some cases, local memory controller **260** or another control component may identify an operating condition associated with a voltage rail (e.g., a

measured or predicted voltage droop or spike), and may respond by coupling a capacitive component with the voltage rail to maintain the voltage on the voltage rail. In some cases, local memory controller **260** (or other control component) may predict a voltage droop or spike on a voltage rail based on receiving a sequence of one or more commands that specify memory operations that may when executed sequentially or in parallel, cause the voltage on a voltage rail to droop or spike.

(62) In some memory architectures, accessing the memory cell **205** may degrade or destroy the logic state stored in a memory cell **205**. For example, a read operation performed in DRAM architectures may partially or completely discharge the capacitor of the target memory cell. The local memory controller **260** may perform a re-write operation or a refresh operation to return the memory cell to its original logic state. The local memory controller **260** may re-write the logic state to the target memory cell after a read operation. In some cases, the re-write operation may be considered part of the read operation. Additionally, activating a single access line, such as a word line **210**, may disturb the state stored in some memory cells in electronic communication with that access line. Thus, a re-write operation or refresh operation may be performed on one or more memory cells that may not have been accessed.

(63) FIG. **3** illustrates an example of a circuit **300** that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein. Circuit **300** may be included in a memory device, such as memory device **110** described with reference to FIG. **1**. Circuit **300** includes an array of memory cells **330** that may be included in a memory array such as memory array **170**, for example.

(64) Circuit **300** may include three voltage rails **305**, each of which may include resistive elements **310** and (optionally) capacitors **315** connected in a grid. Each voltage rail **305** may be coupled with a corresponding supply voltage source via, for example, pad **325** (which may provide electrical access to the voltage rail from off the die or off the memory device) or via a direct electrical connection, using an on-die voltage source. A supply voltage source may be a voltage regulator or power supply, for example, and may be configured to supply (e.g., apply) a substantially constant supply voltage to voltage rail **305**. Voltage rails **305** may be configured to distribute (e.g., transfer) the supply voltage applied by a voltage source to (e.g., across) the array of memory cells **330** and/or to other portions of a memory device using the resistive grid, for example. In some cases, a pad **325** of circuit **300** may be coupled with a V.sub.DD voltage source, a V.sub.SS voltage source, a V.sub.CC voltage source, a V.sub.PP voltage source, or another supply voltage source. Such supply voltage sources may provide power to the array of memory cells **330** and/or to other components in a memory device during operation of the memory device.

(65) In some cases, a supply voltage may be applied to a voltage rail **305** via a pad **325**. Each pad **325** may receive a different supply voltage for the corresponding voltage rail **305**. For example, pad **325-a** may receive a V.sub.SS voltage for voltage rail **305-a**, pad **325-b** may receive a V.sub.DD voltage for voltage rail **305-b**, and/or pad **325-c** may receive a V.sub.CC voltage for voltage rail **305-c**. In some cases, two voltage rails **305** may be coupled with the same voltage source, or two voltage sources may supply the same voltage to two corresponding voltage rails **305**. That is, the voltage rails **305** may be used to distribute the same or different voltages to the array of memory cells **330** and/or to other portions of a memory die.

(66) In some cases, a voltage of voltage rail **305** may vary along voltage rail **305**; that is, the voltage at one location on voltage rail **305** may be different than the voltage at another location on voltage rail **305**. For example, the voltage of voltage rail **305** at a location far away from voltage source may be lower than the voltage of voltage rail **305** nearer to voltage source due to current or voltage demand from various components coupled with voltage rail **305** (such as memory cells in the array of memory cells **330**), or due to parasitic capacitance along voltage rail **305**. Thus, voltage rail **305** may be considered as having various current sources connected to voltage rail **305**, with each current source representing a component (or multiple components) that demands current from

voltage rail **305**.

(67) A voltage rail **305** may include capacitors **315** along voltage rail **305** to help compensate for such voltage variations by supplying additional voltage or charge to the voltage rail **305**. Such capacitors **315** may be hardwired to voltage rail **305** and serve as distributed charge suppliers that may supply additional charge to meet current demand along voltage rail **305**, thereby avoiding propagating such demand back to the voltage source, which may cause a larger voltage droop on the rail.

(68) As described with reference to FIG. **4**, in some cases a memory device may also include a capacitive component that may be dynamically coupled with a voltage rail **305** to maintain a voltage (e.g., a supply voltage) on the voltage rail **305** during periods of high demand.

(69) FIG. **4** illustrates an example of a circuit **400** that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein. Circuit **400** includes voltage rails **305**, pads **325**, and voltage sources as described with reference to FIG. **3**. Such voltage rails **305** may be used to transfer power or voltage from voltage sources to an array of memory cells (such as array of memory cells **330**) and/or to other components on a memory die.

(70) Circuit **400** may include capacitive component **405**. Capacitive component **405** may be dynamically coupled with voltage rail **305-d**, voltage rail **305-e**, and/or voltage rail **305-f** by activating various switching components **415**. That is, capacitive component **405** may be selectively couplable (able to be coupled) with voltage rail **305-d**, voltage rail **305-e**, and/or voltage rail **305-f**. Capacitive component **405** may provide additional capacitance to a voltage rail **305** while it is coupled with the voltage rail **305**. Voltage rail **305** may distribute (e.g., transfer) a voltage to an array of memory cells while it is isolated from capacitive component **405** or while it is coupled with capacitive component **405**. The voltage transferred to the array of memory cells may be based on the supply voltage applied to the voltage rail **305** by the corresponding voltage source. In the ideal case, the voltage transferred to the array of memory cells **330** may be the same as the supply voltage, but in practice the voltage transferred to the array of memory cells **330** may differ from the supply voltage due to voltage droop along voltage rail **305**, for example.

(71) Capacitive component **405** may include a single discrete capacitor. In other cases, capacitive component **405** may include multiple discrete capacitors **420** coupled with each other via a resistive network, as depicted in FIG. **4**. In some cases, a first portion of capacitive component **405** (e.g., a first set of capacitors in capacitive component **405**) may be coupled with one voltage rail **305**, and another portion of capacitive component **405** (e.g., a second set of capacitors in capacitive component **405**) may be coupled with a different voltage rail **305**. That is, some or all of the discrete capacitors **420** in capacitive component **405** may be dynamically allocated to or switched between different voltage rails **305** based on operating conditions associated with the different voltage rails **305**.

(72) In some cases, a control component (such as local memory controller **165**, device memory controller **155**, external memory controller **105**, and/or other control circuitry that may be included on a memory die with the array of memory cells **330**, or a combination thereof) may identify an operating condition associated with a voltage rail **305**, such as a measured or predicted spike or droop in the voltage on the voltage rail **305** or at another location that is coupled with (e.g., receives a voltage from) voltage rail **305**. Based on the identified operating condition, the control component may cause some or all of the capacitive component **405** to be coupled with the voltage rail **305** by activating one or more switching components **415**. Coupling the capacitive component **405** with the voltage rail **305** may help to maintain the target voltage (e.g., the supply voltage) on the voltage rail **305** during some or all of an access operation of the array of memory cells **330**, for example.

(73) In some cases, the operating condition may be identified by sensing operating voltages (e.g., voltages that occur during operation of the memory device) at one or more locations along the voltage rail **305** and/or at one or more locations on a memory device that are coupled with the

voltage rail **305**, and comparing the sensed voltage(s) to a threshold. The voltage(s) may be sensed using one or more sensors, for example. The threshold may be a minimum voltage or a maximum voltage, for example. If one or more of the sensed voltages satisfies the threshold (e.g., a sensed voltage is below a minimum voltage or above a maximum voltage), some or all of capacitive component **405** may be coupled with the voltage rail **305**.

(74) In some cases, instead of comparing each of multiple sensed voltages with a threshold, an average, mean, minimum, maximum, or other function of multiple sensed voltages may be compared with a threshold to identify the operating condition.

(75) The operating condition may be identified by predicting that a voltage at one or more locations along voltage rail **305** and/or at one or more locations that are coupled with voltage rail **305** will satisfy a threshold (e.g., predicting that a voltage will exceed a maximum voltage or droop below a minimum voltage). A control component may receive one or more commands that specify a sequence of one or more memory operations to be performed on the array of memory cells (such as read, write, refresh operations) and predict that a voltage associated with a voltage rail will satisfy the threshold when performing the sequence of one or more memory operations. For example, in a DRAM memory device, a refresh operation may use more power from a V.sub.DD voltage rail and less power from a V.sub.DDQ voltage rail, and thus a control component may predict, based on receiving a command specifying the refresh operation, that the V.sub.DD rail may satisfy a threshold (e.g., that the V.sub.DD rail may need additional capacitance/charge during the refresh operation). In such examples, the control component may couple the capacitive component **405** with the V.sub.DD voltage rail during at least a portion of the refresh operation to maintain the voltage on the V.sub.DD voltage rail.

(76) A control component may predict that a voltage will satisfy the threshold based on previous electrical simulations of the sequence of operations on the memory device, based on or based on predetermined sequences of operations, or using other predictive methods. The commands received by the control component may be pipelined, enabling the control component to predict subsequent operating conditions on the memory die based on the received commands.

(77) A control component may switch some or all of capacitive component **405** (e.g., some or all of the discrete capacitors in capacitive component **405**) from a first voltage rail **305** to a second voltage rail **305** based on an operating condition associated with the second voltage rail **305**. In some cases, the control component may isolate a portion of the capacitive component **405** from the first voltage rail before coupling that portion of the capacitive component **405** with the second voltage rail, by deactivating one or more switching components **415** associated with coupling the portion of the capacitive component **405** with the first voltage rail **305**.

(78) For example, if capacitive component **405** is coupled with voltage rail **305-d** (e.g., switching components **415-a** and **415-b** are activated), the control component may deactivate a first set of switching components (e.g., switching components **415-a** and/or **415-b**) to isolate the capacitive component **405** from voltage rail **305-d** before coupling the capacitive component **405** with voltage rail **305-e** by activating a second set of switching components (e.g., switching components **415-c**, **415-d**, **415-e**, and/or **415-f**).

(79) Capacitive component **405** may be pre-charged to a first pre-charge voltage before capacitive component **405** is coupled with a voltage rail **305** so that capacitive component **405** is capable of providing additional charge or voltage to the voltage rail **305** when capacitive component **405** is coupled with the voltage rail **305**. For example, capacitive component **405** may be pre-charged to a supply voltage associated with the voltage rail **305**, or to a voltage that is higher than the supply voltage associated with the voltage rail **305**, or to another voltage. In some cases, if capacitive component **405** is switched from a first voltage rail (e.g., voltage rail **305-d**) to a second voltage rail (e.g., voltage rail **305-e**), capacitive component **405** may be pre-charged (e.g., recharged) to a second pre-charge voltage before it is coupled with the second voltage rail. The second pre-charge voltage may be the supply voltage associated with the second voltage rail, for example, or another

voltage. In some cases, capacitive component **405** may be pre-charged (recharged) to the second pre-charge voltage after it is isolated from the first voltage rail and before it is coupled with the second voltage rail.

(80) FIG. 5 illustrates an example of a circuit **500** that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein. Circuit **500** may include voltage rails **305**, pads **325**, and voltage sources as described with reference to FIGS. 3 and 4. Such voltage rails **305** may be used to transfer power or voltage from voltage sources to an array of memory cells (such as array of memory cells **330**) and/or to other components on a memory die.

(81) A control component (such as local memory controller **165**, device memory controller **155**, external memory controller **105**, and/or other control circuitry that may be included on a memory die with the array of memory cells **330**) may identify an operating condition associated with a voltage rail **305**, such as a measured or predicted spike or droop in the voltage on the voltage rail **305** or at another location that is coupled with (e.g., receives a voltage from) voltage rail **305**. Based on the identified operating condition, the control component may cause voltage rail **305-g** to be coupled with voltage rail **305-h** by activating one or more switching components **415**. Such coupling of voltage rail **305-g** with voltage rail **305-h** may provide voltage rail **305-g** and/or voltage rail **305-h** with additional capacitance to maintain a voltage on voltage rail **305-g** and/or voltage rail **305-h** during at least a portion of a memory access operation, and may be performed in addition to or instead of coupling a capacitive component with voltage rail **305-g** and/or voltage rail **305-h** as described with reference to FIG. 4.

(82) FIG. 6 shows a block diagram **600** of a memory device **605** that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein. The memory device **605** may be an example of aspects of a memory device as described with reference to FIGS. 1 through 5. The memory device **605** may include a transfer module **610**, an identification module **615**, a coupling module **620**, an isolation module **625**, a charging module **630**, a sensing module **635**, and a command receiving module **640**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(83) The transfer module **610** may transfer, by a voltage rail isolated from a capacitive component, a voltage to an array of memory cells. In some examples, the transfer module **610** may transfer, by the voltage rail coupled with the capacitive component, the voltage to the array of memory cells. In some cases, the capacitive component includes a set of capacitors.

(84) The identification module **615** may identify an operating condition associated with the voltage rail. In some examples, the identification module **615** may identify a second operating condition associated with a second voltage rail for transferring a second voltage to the array of memory cells.

(85) The identification module **615** may identify an operating condition associated with a first voltage rail that is for transferring, based at least in part on a first supply voltage applied to the first voltage rail, a first voltage to an array of memory cells.

(86) The coupling module **620** may couple the capacitive component with the voltage rail based on identifying the operating condition, the capacitive component for maintaining the voltage on the voltage rail for at least a portion of an access operation of the array of memory cells. In some examples, the coupling module **620** may activate a first set of one or more switching components to couple the set of capacitors with the voltage rail. In some examples, the coupling module **620** may activate a second set of one or more switching components to couple at least a portion of the set of capacitors with the second voltage rail. In some examples, the coupling module **620** may couple the portion of the capacitive component with the second voltage rail based on the second operating condition.

(87) The coupling module **620** may couple, based at least in part on identifying the operating condition, the first voltage rail with a second voltage rail for transferring a second voltage to the array of memory cells based at least in part on a second supply voltage applied to the second voltage rail.

(88) The isolation module **625** may isolate a portion of the capacitive component from the voltage rail based on the second operating condition.

(89) The charging module **630** may charge the portion of the capacitive component to a third voltage after isolating the portion of the capacitive component from the voltage rail and before coupling the portion of the capacitive component with the second voltage rail. In some examples, the charging module **630** may charge the capacitive component to a fourth voltage before coupling the capacitive component with the voltage rail. In some cases, the fourth voltage is based on a supply voltage applied to the voltage rail.

(90) The sensing module **635** may sense one or more operating voltages at one or more locations on a die of the array of memory cells. In some cases, identifying the operating condition includes determining that at least one operating voltage of the one or more operating voltages satisfies a threshold.

(91) The command receiving module **640** may receive one or more commands including a sequence of operations to be performed on the array of memory cells. In some cases, identifying the operating condition includes predicting, based on the one or more commands, that the voltage will satisfy a threshold when performing the sequence of operations.

(92) FIG. 7 shows a flowchart illustrating a method or methods **700** that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein. The operations of method **700** may be implemented by a memory device or its components as described herein. For example, the operations of method **700** may be performed by a memory device as described with reference to FIG. 6. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the described functions. Additionally or alternatively, a memory device may perform aspects of the described functions using special-purpose hardware.

(93) At **705**, the memory device may transfer, by a voltage rail isolated from a capacitive component, a voltage to an array of memory cells. The operations of **705** may be performed according to the methods described herein. In some examples, aspects of the operations of **705** may be performed by a transfer component as described with reference to FIG. 6.

(94) At **710**, the memory device may identify an operating condition associated with the voltage rail. The operations of **710** may be performed according to the methods described herein. In some examples, aspects of the operations of **710** may be performed by an identification component as described with reference to FIG. 6.

(95) At **715**, the memory device may couple the capacitive component with the voltage rail based on identifying the operating condition, the capacitive component for maintaining the voltage on the voltage rail for at least a portion of an access operation of the array of memory cells. The operations of **715** may be performed according to the methods described herein. In some examples, aspects of the operations of **715** may be performed by a coupling component as described with reference to FIG. 6.

(96) At **720**, the memory device may transfer, by the voltage rail coupled with the capacitive component, the voltage to the array of memory cells. The operations of **720** may be performed according to the methods described herein. In some examples, aspects of the operations of **720** may be performed by a transfer component as described with reference to FIG. 6.

(97) In some examples, an apparatus as described herein may perform a method or methods, such as the method **700**. The apparatus may include features, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor) for transferring, by a voltage rail isolated from a capacitive component, a voltage to an array of memory cells, identifying an operating condition associated with the voltage rail, coupling the capacitive component with the voltage rail based on identifying the operating condition, the capacitive component for maintaining the voltage on the voltage rail for at least a portion of an access operation of the array of memory cells, and transferring, by the voltage rail coupled with the

capacitive component, the voltage to the array of memory cells.

(98) Some examples of the method **700** and the apparatus described herein may further include operations, features, means, or instructions for identifying a second operating condition associated with a second voltage rail for transferring a second voltage to the array of memory cells, isolating a portion of the capacitive component from the voltage rail based on the second operating condition, and coupling the portion of the capacitive component with the second voltage rail based on the second operating condition.

(99) Some examples of the method **700** and the apparatus described herein may further include operations, features, means, or instructions for charging the portion of the capacitive component to a third voltage after isolating the portion of the capacitive component from the voltage rail and before coupling the portion of the capacitive component with the second voltage rail.

(100) In some examples of the method **700** and the apparatus described herein, the capacitive component includes a set of capacitors. In some examples of the method **700** and the apparatus described herein, coupling the capacitive component with the voltage rail may include operations, features, means, or instructions for activating a first set of one or more switching components to couple the set of capacitors with the voltage rail.

(101) In some examples of the method **700** and the apparatus described herein, coupling the portion of the capacitive component with the second voltage rail may include operations, features, means, or instructions for activating a second set of one or more switching components to couple at least a portion of the set of capacitors with the second voltage rail.

(102) Some examples of the method **700** and the apparatus described herein may further include operations, features, means, or instructions for sensing one or more operating voltages at one or more locations on a die of the array of memory cells, where identifying the operating condition includes determining that at least one operating voltage of the one or more operating voltages satisfies a threshold.

(103) Some examples of the method **700** and the apparatus described herein may further include operations, features, means, or instructions for receiving one or more commands including a sequence of operations to be performed on the array of memory cells, where identifying the operating condition includes predicting, based at least in part on the one or more commands, that the voltage will satisfy a threshold when performing the sequence of operations.

(104) Some examples of the method **700** and the apparatus described herein may further include operations, features, means, or instructions for charging the capacitive component to a fourth voltage before coupling the capacitive component with the voltage rail. In some examples of the method **700** and the apparatus described herein, the fourth voltage may be based on a supply voltage applied to the voltage rail.

(105) FIG. **8** shows a flowchart illustrating a method or methods **800** that supports dynamic allocation of a capacitive component in a memory device in accordance with examples as disclosed herein. The operations of method **800** may be implemented by a memory device or its components as described herein. For example, the operations of method **800** may be performed by a memory device as described with reference to FIG. **6**. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the described functions. Additionally or alternatively, a memory device may perform aspects of the described functions using special-purpose hardware.

(106) At **805**, the memory device may identify an operating condition associated with a first voltage rail that is for transferring, based on a first supply voltage applied to the first voltage rail, a first voltage to an array of memory cells. The operations of **805** may be performed according to the methods described herein. In some examples, aspects of the operations of **805** may be performed by an identification module as described with reference to FIG. **6**.

(107) At **810**, the memory device may couple, based on identifying the operating condition, the first voltage rail with a second voltage rail for transferring a second voltage to the array of memory

cells based on a second supply voltage applied to the second voltage rail. The operations of **810** may be performed according to the methods described herein. In some examples, aspects of the operations of **810** may be performed by a coupling module as described with reference to FIG. 6. (108) In some examples, an apparatus as described herein may perform a method or methods, such as the method **800**. The apparatus may include features, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor) for identifying an operating condition associated with a first voltage rail that is for transferring, based on a first supply voltage applied to the first voltage rail, a first voltage to an array of memory cells and coupling, based on identifying the operating condition, the first voltage rail with a second voltage rail for transferring a second voltage to the array of memory cells based on a second supply voltage applied to the second voltage rail.

(109) Some examples of the method **800** and the apparatus described herein may further include operations, features, means, or instructions for sensing one or more operating voltages at one or more locations on a die of the array of memory cells, where identifying the operating condition includes determining that at least one operating voltage of the one or more operating voltages satisfies a threshold.

(110) Some examples of the method **800** and the apparatus described herein may further include operations, features, means, or instructions for receiving one or more commands including a sequence of operations to be performed on the array of memory cells, where identifying the operating condition includes predicting, based at least in part on the one or more commands, that the first voltage will satisfy a threshold when performing the sequence of operations.

(111) In some examples of the method **800** and the apparatus described herein, coupling the second voltage rail with the first voltage rail may include operations, features, means, or instructions for activating a set of one or more switching components to couple the second voltage rail with the first voltage rail.

(112) It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, aspects from two or more of the methods may be combined.

(113) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, it will be understood by a person of ordinary skill in the art that the signal may represent a bus of signals, where the bus may have a variety of bit widths.

(114) As used herein, the term “virtual ground” refers to a node of an electrical circuit that is held at a voltage of approximately zero volts (0V) but that is not directly coupled with ground. Accordingly, the voltage of a virtual ground may temporarily fluctuate and return to approximately 0V at steady state. A virtual ground may be implemented using various electronic circuit elements, such as a voltage divider consisting of operational amplifiers and resistors. Other implementations are also possible. “Virtual grounding” or “virtually grounded” means connected to approximately 0V.

(115) The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (or in conductive contact with or connected with or coupled with) one another if there is any conductive path between the components that can, at any time, support the flow of signals between the components. At any given time, the conductive path between components that are in electronic communication with each other (or in conductive contact with or connected with or coupled with) may be an open

circuit or a closed circuit based on the operation of the device that includes the connected components. The conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some cases, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

(116) The term “coupling” refers to condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components over a conductive path to a closed-circuit relationship between components in which signals can be communicated between components over the conductive path. When a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.

(117) The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other when the switch is open. When a controller isolates two components from one another, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.

(118) As used herein, the term “substantially” means that the modified characteristic (e.g., a verb or adjective modified by the term substantially) need not be absolute but is close enough to achieve the advantages of the characteristic.

(119) The devices discussed herein, including a memory array, may be formed on a semiconductor substrate, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some cases, the substrate is a semiconductor wafer. In other cases, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

(120) A switching component or a transistor discussed herein may represent a field-effect transistor (FET) and comprise a three terminal device including a source, drain, and gate. The terminals may be connected to other electronic elements through conductive materials, e.g., metals. The source and drain may be conductive and may comprise a heavily-doped, e.g., degenerate, semiconductor region. The source and drain may be separated by a lightly-doped semiconductor region or channel. If the channel is n-type (i.e., majority carriers are electrons), then the FET may be referred to as a n-type FET. If the channel is p-type (i.e., majority carriers are holes), then the FET may be referred to as a p-type FET. The channel may be capped by an insulating gate oxide. The channel conductivity may be controlled by applying a voltage to the gate. For example, applying a positive voltage or negative voltage to an n-type FET or a p-type FET, respectively, may result in the channel becoming conductive. A transistor may be “on” or “activated” when a voltage greater than or equal to the transistor's threshold voltage is applied to the transistor gate. The transistor may be “off” or “deactivated” when a voltage less than the transistor's threshold voltage is applied to the transistor gate.

(121) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an

example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.

(122) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

(123) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(124) The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(125) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(126) The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method, comprising: coupling a first portion of a capacitive component with a first voltage rail based at least in part on a first operating condition, wherein the first voltage rail is coupled with an array of memory cells; transferring, by the first voltage rail coupled with the first portion of the capacitive component, a first voltage to the array of memory cells; coupling a second portion of the capacitive component with a second voltage rail based at least in part on a second operating condition, wherein the second voltage rail is coupled with the array of memory cells; and transferring, by the second voltage rail coupled with the second portion of the capacitive component, a second voltage to the array of memory cells, wherein the capacitive component comprises a plurality of sets of capacitors coupled with each other via a resistive network, and wherein the first portion of the capacitive component is associated with a first set of capacitors of the plurality of sets of capacitors and the second portion of the capacitive component is associated with a second set of capacitors of the plurality of sets of capacitors.
2. The method of claim 1, further comprising: identifying the first operating condition associated with the first voltage rail, wherein coupling the first portion of the capacitive component with the first voltage rail is based at least in part on identifying the first operating condition; and identifying the second operating condition associated with the second voltage rail, wherein coupling the second portion of the capacitive component with the second voltage rail is based at least in part on identifying the second operating condition.
3. The method of claim 1, further comprising: coupling a third portion of the capacitive component with a third voltage rail based at least in part on a third operating condition, wherein the third voltage rail is coupled with the array of memory cells; and transferring, by the third voltage rail coupled with the third portion of the capacitive component, a third voltage to the array of memory cells.
4. The method of claim 1, further comprising: coupling a third portion of the capacitive component with the first voltage rail, the second voltage rail, and a third voltage rail based at least in part on a third operating condition associated with the first voltage rail, the second voltage rail, the third voltage rail, or any combination thereof.
5. The method of claim 1, further comprising: activating one or more first switching components associated with the first portion of the capacitive component and the first voltage rail, wherein coupling the first portion of the capacitive component with the first voltage rail is based at least in part on activating the one or more first switching components; and activating one or more second switching components associated with the second portion of the capacitive component and the second voltage rail, wherein coupling the second portion of the capacitive component with the second voltage rail is based at least in part on activating the one or more second switching components.
6. The method of claim 1, further comprising: identifying a third operating condition associated with the second voltage rail; and coupling the first portion of the capacitive component with the second voltage rail based at least in part on the third operating condition.
7. The method of claim 6, further comprising: deactivating one or more first switching components associated with the first portion of the capacitive component and the first voltage rail; and activating one or more second switching components associated with the first portion of the capacitive component and the second voltage rail, wherein coupling the first portion of the capacitive component with the second voltage rail is based at least in part on activating the one or more second switching components.
8. An apparatus, comprising: an array of memory cells; a first voltage rail coupled with the array of memory cells and configured to transfer a first voltage to the array of memory cells; a second voltage rail coupled with the array of memory cells and configured to transfer a second voltage to the array of memory cells; a capacitive component comprising a plurality of sets of capacitors; a first set of switching components configured to dynamically couple one or more first sets of

capacitors of the plurality of sets of capacitors in the capacitive component with the first voltage rail based at least in part on a first operating condition associated with the first voltage rail; and a second set of switching components configured to dynamically couple one or more second sets of capacitors of the plurality of sets of capacitors in the capacitive component with the second voltage rail based at least in part on a second operating condition associated with the second voltage rail.

9. The apparatus of claim 8, further comprising: a control component configured to identify the first operating condition associated with the first voltage rail and the second operating condition associated with the second voltage rail, wherein the first set of switching components is configured to dynamically couple the one or more first sets of capacitors to the first voltage rail based at least in part on the control component identifying the first operating condition and the second set of switching components is configured to dynamically couple the one or more second sets of capacitors to the second voltage rail based at least in part on identifying the second operating condition.

10. The apparatus of claim 8, further comprising: a third voltage rail coupled with the array of memory cells and configured to transfer a third voltage to the array of memory cells; and a third set of switching components configured to dynamically couple each set of capacitors of the plurality of sets of capacitors in the capacitive component with the third voltage rail based at least in part on a third operating condition associated with the third voltage rail.

11. The apparatus of claim 8, further comprising: a third set of switching components configured to dynamically couple each set of capacitors of the plurality of sets of capacitors of the capacitive component with the first voltage rail, the second voltage rail, and a third voltage rail based at least in part on a third operating condition associated with the first voltage rail, the second voltage rail, the third voltage rail, or any combination thereof.

12. The apparatus of claim 8, further comprising: a control component configured to identify a third operating condition associated with the second voltage rail; and a third set of switching components configured to dynamically decouple a first set of capacitors of the plurality of sets of capacitors of the capacitive component from the first voltage rail and couple the first set of capacitors with the second voltage rail based at least in part on the third operating condition.

13. The apparatus of claim 12, wherein the third set of switching components comprises the first set of switching components and the second set of switching components.

14. A memory system, comprising: one or more memory devices; and circuitry coupled with the one or more memory devices and configured to cause the memory system to: couple a first portion of a capacitive component with a first voltage rail based at least in part on a first operating condition, wherein the first voltage rail is coupled with an array of memory cells; transfer, by the first voltage rail coupled with the first portion of the capacitive component, a first voltage to the array of memory cells; couple a second portion of the capacitive component with a second voltage rail based at least in part on a second operating condition, wherein the second voltage rail is coupled with the array of memory cells; and transfer, by the second voltage rail coupled with the second portion of the capacitive component, a second voltage to the array of memory cells, wherein the capacitive component comprises a plurality of sets of capacitors coupled with each other via a resistive network, and wherein the first portion of the capacitive component is associated with a first set of capacitors of the plurality of sets of capacitors and the second portion of the capacitive component is associated with a second set of capacitors of the plurality of sets of capacitors.

15. The memory system of claim 14, wherein the circuitry is further configured to cause the memory system to: identify the first operating condition associated with the first voltage rail, wherein coupling the first portion of the capacitive component with the first voltage rail is based at least in part on identifying the first operating condition; and identify the second operating condition associated with the second voltage rail, wherein coupling the second portion of the capacitive component with the second voltage rail is based at least in part on identifying the second operating condition.

16. The memory system of claim 14, wherein the circuitry is further configured to cause the memory system to: couple a third portion of the capacitive component with a third voltage rail based at least in part on a third operating condition, wherein the third voltage rail is coupled with the array of memory cells; and transfer, by the third voltage rail coupled with the third portion of the capacitive component, a third voltage to the array of memory cells.

17. The memory system of claim 14, wherein the circuitry is further configured to cause the memory system to: couple a third portion of the capacitive component with the first voltage rail, the second voltage rail, and a third voltage rail based at least in part on a third operating condition associated with the first voltage rail, the second voltage rail, the third voltage rail, or any combination thereof.

18. The memory system of claim 14, wherein the circuitry is further configured to cause the memory system to: activate one or more first switching components associated with the first portion of the capacitive component and the first voltage rail, wherein coupling the first portion of the capacitive component with the first voltage rail is based at least in part on activating the one or more first switching components; and activate one or more second switching components associated with the second portion of the capacitive component and the second voltage rail, wherein coupling the second portion of the capacitive component with the second voltage rail is based at least in part on activating the one or more second switching components.

19. The memory system of claim 14, wherein the circuitry is further configured to cause the memory system to: identify a third operating condition associated with the second voltage rail; and couple the first portion of the capacitive component with the second voltage rail based at least in part on the third operating condition.
